

SUMMARY

Currently, a second-year Ph.D. candidate in Statistics at North Carolina State University, specializing in advanced statistical modeling and machine learning applications. Experienced in Python and R programming with a proven track record of translating complex data challenges into actionable insights through sophisticated statistical models. Actively seeking a Summer 2025 internship to leverage expertise in statistical analysis and machine learning while contributing to impactful research initiatives.

EDUCATION

Degree	Institute	CGPA/GPA	Year
Ph.D. in Statistics	North Carolina State University	3.88/4	2023-27(Expected)
M.Sc. Applied Statistics & Informatics	Indian Institute of Technology, Bombay	8.88/10	2019-21
B.Sc. Statistics (Minor: Mathematics, Economics)	University of Mumbai	9.72/10	2016-19

WORK EXPERIENCE

- Wells Fargo | Quantitative Analytics Specialist (Model Validation)

Corporate Model Risk Team | Bengaluru, India

Jul 2021 – Jul 2023

- Executed end-to-end model validation for credit risk and loss forecasting models per SR 11-7 guidelines, performing statistical testing and documentation review
 - Developed a network-based machine learning algorithm for model risk monitoring and early warning system; achieved finalist position in Corporate Risk Innovation Challenge 2022
 - Led winning project in Wells Fargo Hackathon, using XGBoost & time series analysis to predict call center volumes
 - Received Manager’s Spotlight Award for consistent high performance
- Accenture Operations |Data Science Intern

Insights & Intelligence Unit | Bengaluru, India

Apr 2020 – Jun 2020

- Developed ML models to predict advertisement click-through rates on a high-cardinality web dataset
 - Reduced dimensionality and improved model performance by applying target encoding, one-hot encoding, TF-IDF, and PCA techniques to complex categorical and textual features
 - Outperformed baseline logistic regression by optimizing XGBoost hyperparameters, increasing AUC from 0.52 to 0.72
 - Awarded a pre-placement offer based on exceptional contributions and project outcomes during the internship
- Cytel Statistical Software & Services Pvt. Ltd.|Project Trainee

Data Science Unit | Pune, India

May 2019 – Jun 2019

- Conducted in-depth exploratory data analysis on heart disease dataset, identifying key risk factors and leveraging insights to guide feature selection for predictive modeling
 - Developed and optimized machine learning models (Random Forest, XGBoost, Logistic Regression) in R, achieving 87% accuracy in predicting heart disease risk, demonstrating strong data science and healthcare analytics skills
- Data Analytics and Visualization Team, IIT Bombay |Head

UG Academic Council, IIT Bombay | Mumbai, India

May 2020 – Apr 2021

- Led an 8-member team in data-driven projects, delivering key insights through comprehensive reports and impactful initiatives
 - Designed & analyzed student surveys that directly shaped the Institute’s COVID-19 grading policy, positively impacting 10K students
 - Leveraged data visualization (Google Data Studio) to provide actionable insights on student feedback and online learning readiness

PROJECTS

- Time Series Analysis of COVID-19 Data

- Developed and implemented predictive models combining time series analysis (Autoregressive, DACP) and machine learning classifiers (Logistic Regression, SVM, Decision Trees) to forecast COVID-19 trends and patient outcomes, enabling data-driven pandemic management decisions for the Pune Municipal Corporation.
- Causal Inference for Social Network Data | Course Project

- Explored semiparametric methods for causal inference in social networks, incorporating both information transmission across ties and latent similarities among connected nodes, and demonstrated their effectiveness through simulation studies.
- Demand Prediction for Sports Goods | Summer RA Project

- Developed a demand forecasting model to predict SKU-level demand across brands and time horizons. Incorporated data from multiple platforms to account for SKU-specific seasonal patterns, improving forecasting precision and business insights.
- Predicting Permanent Magnet (PM) Temperature of Motor using Regression | Course Project

- Performed comprehensive data analysis to identify heteroscedasticity, autocorrelation, outliers, and non-normality of errors; addressed multicollinearity using Variance Inflation Factor (VIF) and Principal Component Analysis (PCA); and optimized predictor selection through a stepwise approach based on Adjusted R², AIC, and BIC.

RELEVANT COURSEWORK

- Statistics:

Regression Analysis, Statistical Inference, Data Structures, Statistical Techniques in Data Mining, Bayesian Inference, Linear Algebra, Computation for Statistical Research, Introduction to Derivative Pricing, Linear Algebra, Combinatorics, Optimization, Causal Inference
- Technical Skills:

Python, R, C++, SQL (basic), SAS (basic),LaTeX

EXTRACURRICULAR ACTIVITIES

- Mentored undergraduate students at the NCSU-Duke Summer Institute of Biostatistics (SIBS) 2024, fostering interest in statistics and biostatistics through field visits and providing academic guidance
 - Silver medallist in 57th National School Gymnastics Championship held at Kolkata, West Bengal